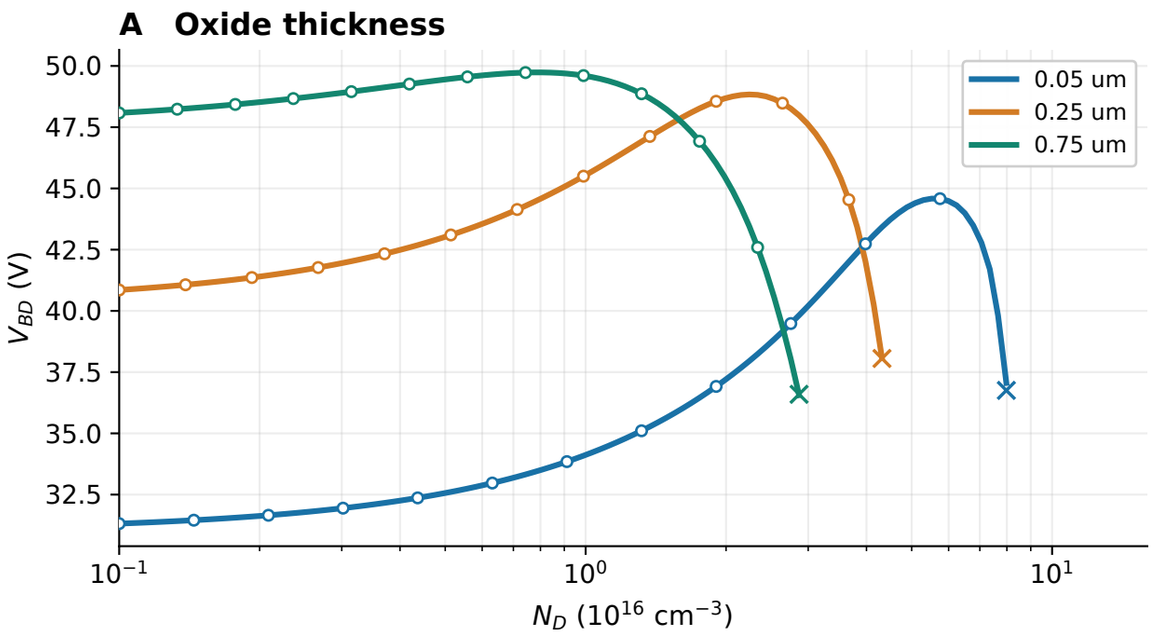
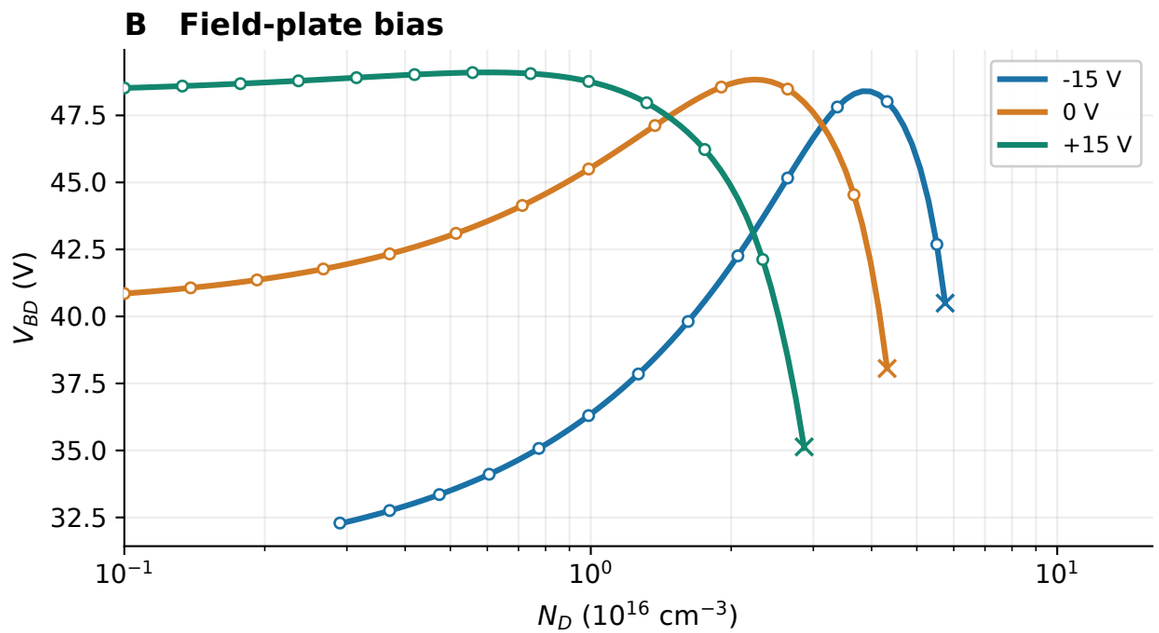
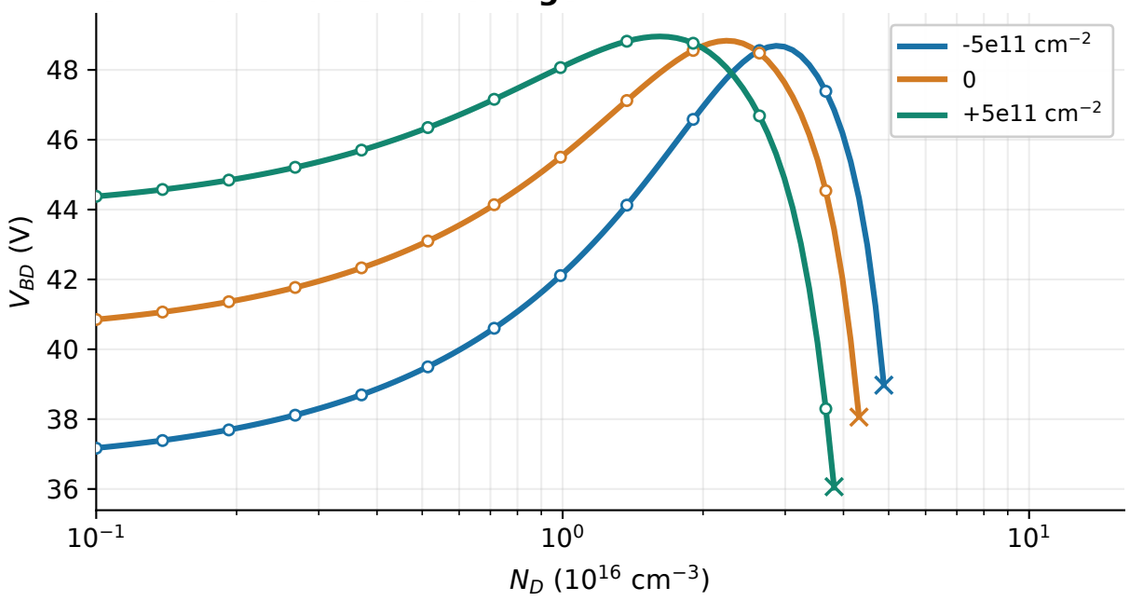
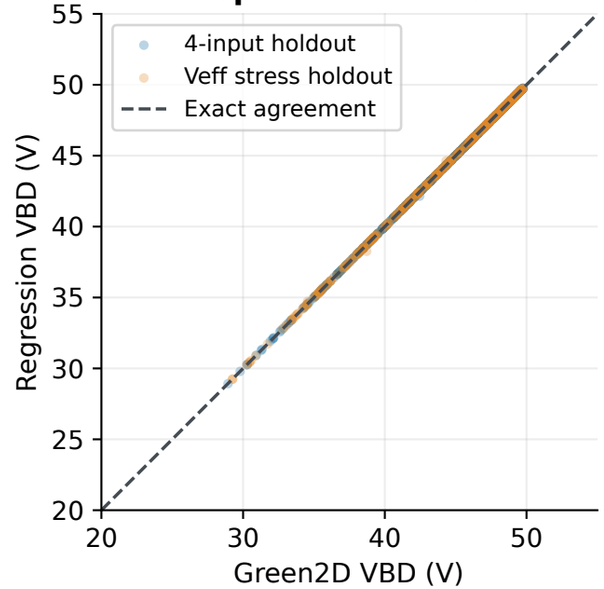
Four-input breakdown regression | 2D Green-function reference

Inputs: N_D, t_{ox}, V_{FP} and N_{ox} | Fixed: silicon half-width a = 0.5 um; drift length L = 1.6 um; 300 K







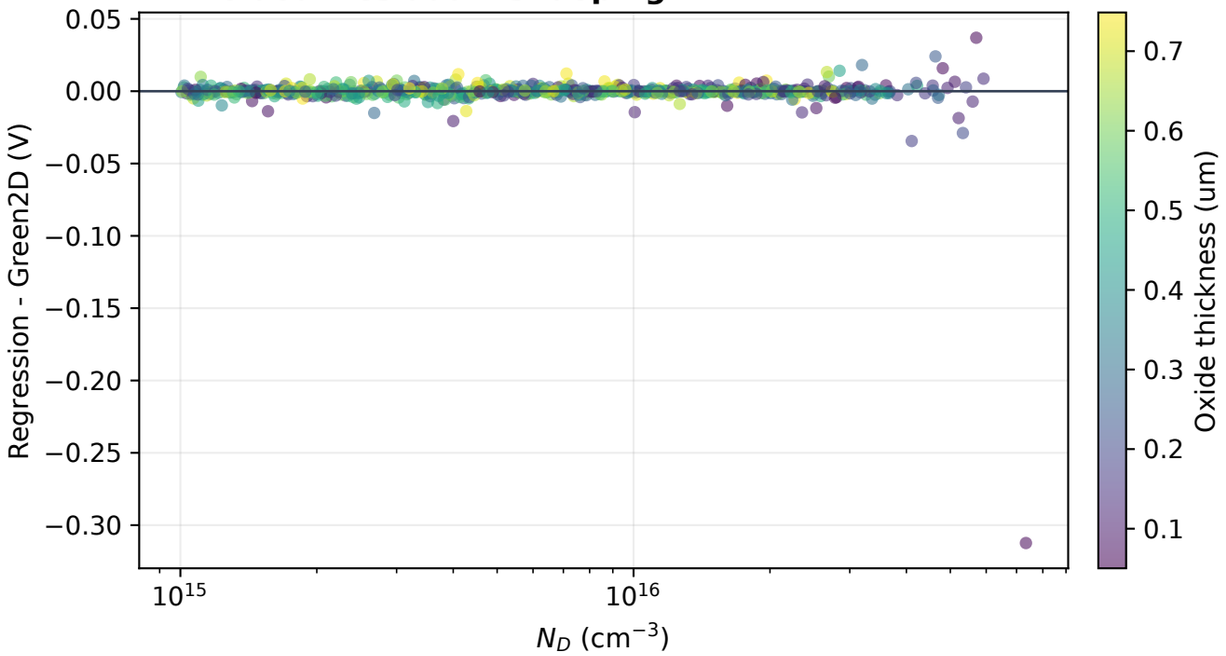


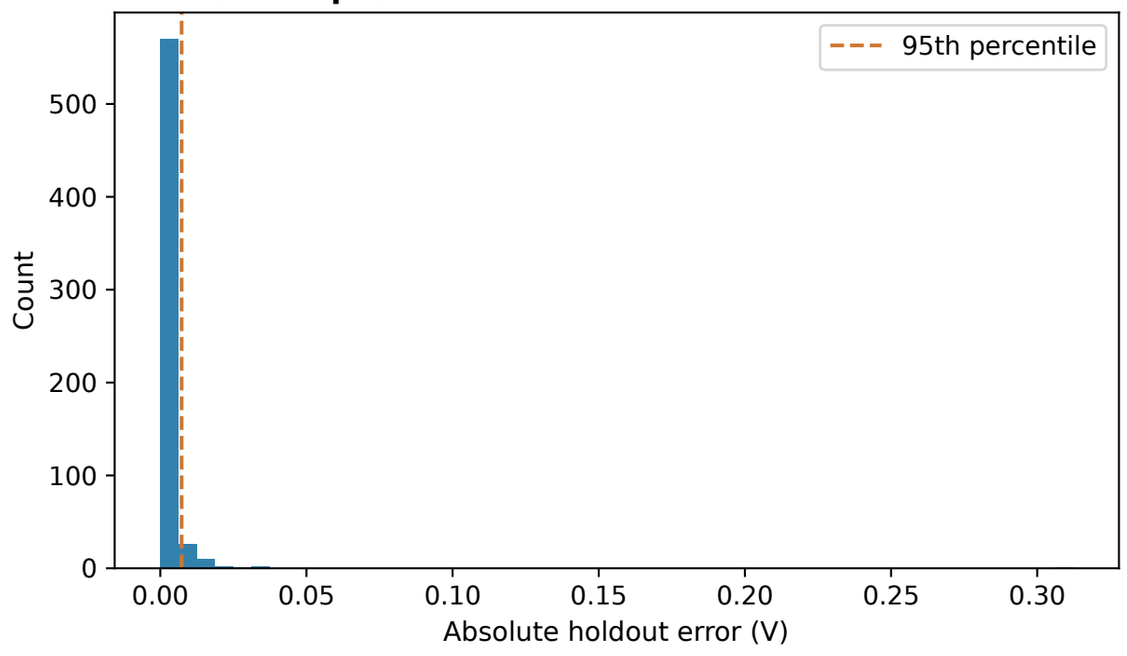
Lines: polynomial + local RBF residual regression | Open circles: Green2D ionization roots | x: last admissible computed point

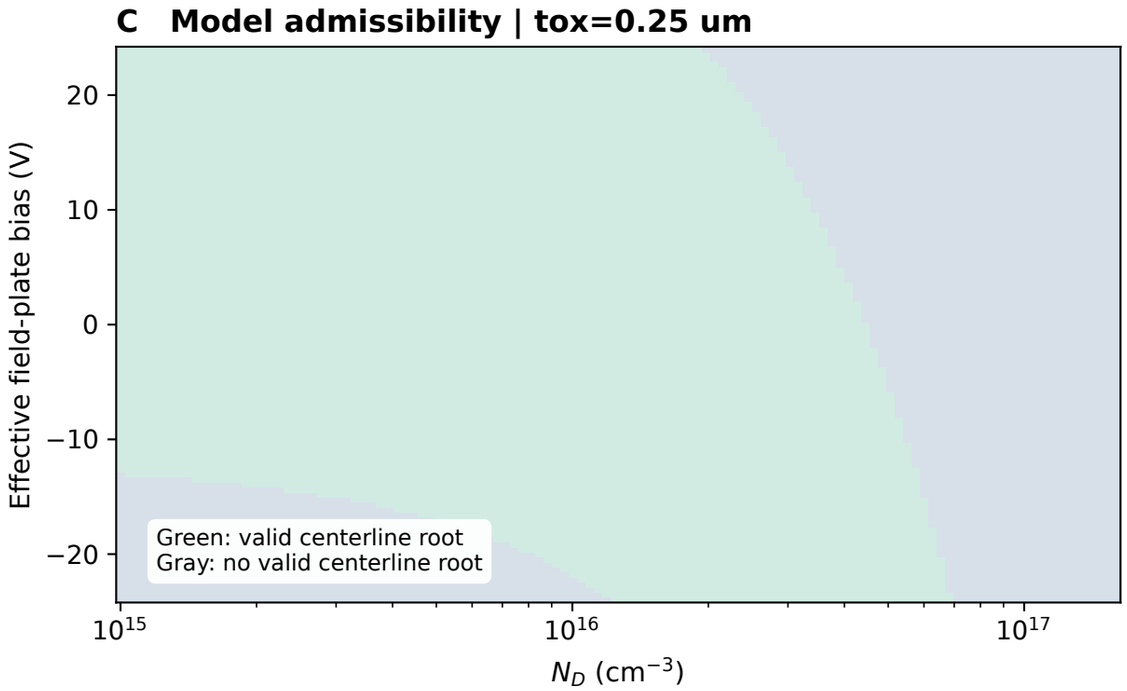
No invalid-domain extrapolation. Model is centerline avalanche in a fully depleted rectangular benchmark; not calibrated TCAD.

Validation and admissibility of the four-input regression

Test sets are off-grid. Invalid inputs are never used as numerical zeroes or infinite breakdown voltages.








```
HOLDOUT VALIDATION  |  Independent 4-input test
Valid 612 of 900 independent samples
R-squared: 0.9999913
RMSE: 0.0133 V
MAE: 0.0027 V
95th percentile absolute error: 0.0073 V
Maximum absolute error: 0.3124 V

ADDITIONAL EFFECTIVE-BIAS STRESS TEST
Valid 470 of 700 samples
RMSE: 0.0276 V
Maximum error: 0.4632 V

NB: Invalid domain is checked using the Green model,
not assigned infinite or extrapolated VBD.
```